Alkylphenols--potential modulators of the allergic response.

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The prevalence of allergic diseases has increased in recent decades. Allergic diseases, particularly asthma, are complex diseases with strong gene-environment interactions. Epidemiological studies have identified a variety of risk factors for the development of allergic diseases. Among them, endocrine-disrupting chemicals (EDCs) play an important role in triggering or exacerbating these diseases. 4-Nonylphenol (NP) and 4-octylphenol (OP)--two major alkylphenols--have been recognized as common toxic and xenobiotic endocrine disrupters. Due to their low solubility, high hydrophobicity, and low estrogenic activity, they tend to accumulate in the human body and may be associated with the adverse effects of allergic diseases. Recently, new evidence has supported the importance of alkylphenols in the in vitro allergic response. This review focuses on the effects of alkylphenols on several key cell types in the context of allergic inflammation. Copyright © 2012. Published by Elsevier B.V.